

Exercise 4.1

Prove

$$\prod_{i=1}^n \frac{1}{A_i^{\alpha_i}} = \frac{\Gamma(\alpha)}{\prod_{i=1}^n \Gamma(\alpha_i)} \left[\prod_{i=1}^n \int_0^1 dx_i x_i^{\alpha_i-1} \right] \frac{\delta(1-x)}{\left[\sum_{i=1}^n x_i A_i \right]^\alpha}$$

where $\alpha_i \in \mathbb{R}$ and $\text{Re}(\alpha_i) > 0$

$$\alpha = \sum_{i=1}^n \alpha_i \quad \text{and} \quad x = \sum_{i=1}^n x_i$$

The Gamma function is defined as $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$, $\text{Re}(z) > 0$ Let's start with using $\frac{1}{A^\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty dt t^{\alpha-1} e^{-tA}$, proof of identity:

$$\int_0^\infty dt t^{\alpha-1} e^{-tA} \rightarrow \text{change of variables } u = tA \rightarrow \int_0^\infty du A^{-1} (uA^{-1})^{\alpha-1} e^{-u} = \int_0^\infty du u^{\alpha-1} e^{-u} \frac{A^{-1}}{A^{\alpha-1}} = \frac{\Gamma(\alpha)}{A^\alpha}$$

We can easily generalize that to

Scaling property of delta-function

$$\prod_{i=1}^n \left[\frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} \right] = \prod_{i=1}^n \left(\int_0^\infty dt_i t_i^{\alpha_i-1} e^{-t_i A_i} \right), \quad \text{Now we insert } 1 = \int_0^\infty dk \delta(k - \sum t_i) = \int_0^\infty k^{-1} \delta(1 - \frac{\sum t_i}{k}) dk$$

$$\prod_{i=1}^n \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \prod_{i=1}^n \left[\int_0^\infty \int_0^\infty dt_i dk t_i^{\alpha_i-1} k^{-1} \delta(1 - \frac{\sum t_i}{k}) e^{-t_i A_i} \right]$$

Now let's do change of variables: $\frac{t_i}{k} = z_i \Rightarrow \frac{\sum t_i}{k} = \sum z_i = 1$
 $\Rightarrow dt_i = dz_i k$

$$= \prod_{i=1}^n \left[\int_0^\infty \int_0^\infty dz_i k dk (z_i k)^{\alpha_i-1} k^{-1} \delta(1-z) e^{-k(z_i A_i)} \right]$$

$$= \prod_{i=1}^n \left[\int_0^\infty dz_i z_i^{\alpha_i-1} \delta(1-z) \int_0^\infty dk k^{\alpha_i-1} e^{-k(z_i A_i)} \right]$$

$$= \left[\prod_{i=1}^n \int_0^\infty dz_i z_i^{\alpha_i-1} \right] \delta(1-z) \underbrace{\int_0^\infty dk k^{\alpha-1} e^{-k(\sum z_i A_i)}}_{= \frac{\Gamma(\alpha)}{(\sum z_i A_i)^\alpha}}$$

$\prod_{i=1}^n k^{\alpha_i-1} = k^{\sum \alpha_i - 1} = k^{\alpha-1}$
 $\int_0^\infty dt t^{\alpha-1} e^{-kA} = \frac{\Gamma(\alpha)}{A^\alpha}$

Since we have delta-function $\delta(1-z)$ and integration range of z_i starts at $z_i = 0$ the maximum range for integral is actually 1 because $z_i > 1$ give 0 contribution because of δ -function

Let's also rename $z \rightarrow x$
 $z_i \rightarrow x_i$

$$\prod_{i=1}^n \frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \left(\prod_{i=1}^n \int_0^\infty dz_i z_i^{\alpha_i-1} \right) \delta(1-z) \frac{\Gamma(\alpha)}{(\sum z_i A_i)^\alpha}$$

$$\prod_{i=1}^n \frac{1}{A_i^{\alpha_i}} = \frac{\Gamma(\alpha)}{\prod_{i=1}^n \Gamma(\alpha_i)} \left[\prod_{i=1}^n \int_0^1 dx_i x_i^{\alpha_i-1} \right] \frac{\delta(1-x)}{(\sum x_i A_i)^\alpha}$$

Exercise 4.2

$$\mathcal{I} = \mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^4}$$

Prove $\mathcal{I} = 0$;

$$\mathcal{I} = \mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{k^2 - m^2}{k^4 (k^2 - m^2)} = \mu^\epsilon \left[\int \frac{d^D k}{(2\pi)^D} \frac{k^2}{k^4 (k^2 - m^2)} - \frac{m^2}{k^4 (k^2 - m^2)} \right]$$

We can see that $\int \frac{d^D k}{(2\pi)^D} \frac{k^2}{k^4 (k^2 - m^2)} \sim \int \frac{k^{D+1} dk}{k^4 (k^2 - m^2)}$

This has following potential divergencies: $\int \frac{k^5}{k^4 (k^2 - m^2)} dk = \int \frac{k}{k^2 - m^2} \stackrel{UV}{\sim} \ln(k^2 - m^2)$

IR-divergence for $D < 3$

UV-divergence for $D \geq 4$ so let's write for this integral $D = 4 - \epsilon_{UV}$ ($\epsilon_{UV} > 0$)

Now let's check for $\int \frac{d^D k}{(2\pi)^D} \frac{m^2}{k^4 (k^2 - m^2)} \sim \int \frac{k^{D-1} dk}{k^4 (k^2 - m^2)}$

IR-divergence for $D \leq 4$

$\Rightarrow$ let's write now $D = 4 - \epsilon_{IR}$

UV-divergence for $D \geq 6$ so no UV-divergence near $D=4$

Now let's use known results to calculate our UV- and IR-divergent Integrals

$$\begin{aligned} \mathcal{I}_{UV} &= \int \frac{d^D k}{(2\pi)^D} \frac{k^2}{k^4 (k^2 - m^2)} \quad \text{let's now Wick-rotate } d^D k \rightarrow d^D k_E \quad k^2 = -k_E^2, \quad k^4 = k_E^4 \\ &= \int \frac{d^D k_E}{(2\pi)^D} \frac{k_E^2}{k_E^4 (k_E^2 + m^2)} = \int \frac{d^D k_E}{(2\pi)^D} \int d\Omega_D \frac{k_E^{D+1}}{k_E^4 (k_E^2 + m^2)} \\ &= \frac{2 \pi^{D/2}}{(2\pi)^D \Gamma(D/2)} \int dl \frac{l^{1-\epsilon}}{(l^2 + m^2)} \end{aligned}$$

Peskin (7.81):
 $\int d\Omega_D = \frac{2 \pi^{D/2}}{\Gamma(D/2)}$
 let's rename k_E to l
 $D = 4 - \epsilon_{UV} \equiv 4 - \epsilon$

Now let's use the known result for this integral:

$$\int dl \frac{l^a}{(l^2 + \Delta)^b} = \Delta^{\frac{a+1}{2}-b} \frac{\Gamma(\frac{a+1}{2}) \Gamma(b - \frac{a+1}{2})}{2 \Gamma(b)}$$

Schwartz eq (B.35)
 $\Gamma(1) = 1$

We have $a = 1 - \epsilon$ and $b = 1$:

$$\mathcal{I}_{UV} = \frac{2 \pi^{D/2}}{(2\pi)^D} (m^2)^{\frac{1-\epsilon+1}{2}-1} \frac{\Gamma(\frac{1-\epsilon+1}{2}) \Gamma(1 - \frac{1-\epsilon+1}{2})}{2 \Gamma((4-\epsilon)/2)}$$

$$= \frac{\pi^{D/2}}{(2\pi)^D} (m^2)^{-\frac{\epsilon}{2}} \Gamma(1 - \frac{\epsilon}{2}) \Gamma(\frac{\epsilon}{2}) \Gamma^{-1}(2 - \frac{\epsilon}{2})$$

let's now use $\Gamma(1+x) = x \Gamma(x)$
 to get $\Gamma(1 - \frac{\epsilon}{2}) = -\frac{\epsilon}{2} \Gamma(-\frac{\epsilon}{2})$

So the UV-divergent part is

$$I_{UV} = \frac{\pi^{D/2}}{(2\pi)^D} (m^2)^{-\epsilon/2} \left[\frac{-\epsilon/2 \Gamma(-\epsilon/2) \Gamma(\epsilon/2)}{\Gamma(2-\epsilon/2)} \right]$$

Now let's evaluate the IR-divergent integral:

$$\begin{aligned} I_{IR} &= - \int \frac{m^2}{k^4(k^2-m^2)} \frac{d^D k}{(2\pi)^D} \stackrel{\text{Wick rotate}}{=} \frac{+m^2}{(2\pi)^D} \int d\Omega_D \int dl \frac{l^{D-1}}{l^4(l^2+m^2)} \quad \parallel \int d\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)} \\ &= + \frac{2\pi^{D/2}}{(2\pi)^D} \frac{m^2}{\Gamma(D/2)} \int dl \frac{l^{D-1}}{l^4(l^2+m^2)} \quad \parallel \text{let's now write } D=4-\epsilon_{IR} \equiv 4-\epsilon \\ &= + \frac{2\pi^{D/2}}{(2\pi)^D} \frac{m^2}{\Gamma(2-\epsilon/2)} \int dl \frac{l^{-\epsilon}}{l(l^2+m^2)} = \frac{+2\pi^{D/2}}{(2\pi)^D} \frac{m^2}{\Gamma(2-\epsilon/2)} \int dl \frac{l^{-1-\epsilon}}{(l^2+m^2)} \end{aligned}$$

Now let's use Schwartz eq. (B.35) again

$$\begin{aligned} \int dl \frac{l^{-1-\epsilon}}{(l^2+m^2)} &= (m^2)^{\frac{-1-\epsilon+1}{2}-1} \frac{\Gamma(\frac{-1-\epsilon+1}{2}) \Gamma(1-\frac{-1-\epsilon+1}{2})}{2 \Gamma(1)} \\ \Rightarrow I_{IR} &= \frac{m^2 \pi^{D/2}}{(2\pi)^D} (m^2)^{-1-\epsilon/2} \left[\frac{\Gamma(-\epsilon/2) \Gamma(1+\epsilon/2)}{\Gamma(2-\epsilon/2)} \right] \quad \parallel \begin{aligned} \Gamma(1+\epsilon/2) &= \epsilon/2 \Gamma(\epsilon/2) \\ m^2 (m^2)^{-1-\epsilon/2} &= (m^2)^{-\epsilon/2} \end{aligned} \\ \Rightarrow I_{IR} &= \frac{\pi^{D/2}}{(2\pi)^D} m^{-\epsilon/2} \left[\frac{\epsilon/2 \Gamma(\epsilon/2) \Gamma(-\epsilon/2)}{\Gamma(2-\epsilon/2)} \right] \end{aligned}$$

Now again showing $D = D_{IR} = 4 - \epsilon_{IR}$ and $D = D_{UV} = 4 - \epsilon_{UV}$ for I_{IR} and I_{UV} we can write full integral I as

$$I = (I_{IR} + I_{UV}) \mu^\epsilon = \mu^\epsilon \left[\frac{\pi^{D_{IR}/2}}{(2\pi)^{D_{IR}}} (m^2)^{-\frac{\epsilon_{IR}}{2}} \left(\frac{\frac{\epsilon_{IR}}{2} \Gamma(\frac{\epsilon_{IR}}{2}) \Gamma(-\frac{\epsilon_{IR}}{2})}{\Gamma(2-\epsilon_{IR}/2)} \right) + \frac{\pi^{D_{UV}/2}}{(2\pi)^{D_{UV}}} (m^2)^{-\frac{\epsilon_{UV}}{2}} \left(\frac{-\epsilon_{UV} \Gamma(-\epsilon_{UV}/2) \Gamma(\epsilon_{UV}/2)}{2 \Gamma(2-\epsilon_{UV}/2)} \right) \right]$$

Now let's set $\epsilon_{UV} = \epsilon_{IR}$ since ϵ_{UV} and ϵ_{IR} are small and go to 0 to get $D=4 = D_{IR} = D_{UV}$
 $\Rightarrow D_{IR} = D_{UV} = D$ and $\epsilon_{UV} = \epsilon_{IR} = \epsilon$

$$\Rightarrow I = \mu^\epsilon \frac{\pi^{D/2}}{(2\pi)^D} (m^2)^{-\frac{\epsilon}{2}} \frac{\epsilon_{IR}}{2} \frac{\Gamma(\epsilon_{IR}/2) \Gamma(-\frac{\epsilon_{IR}}{2})}{\Gamma(2-\epsilon_{IR}/2)} \cdot \underbrace{[1-1]}_{=0} = 0$$

So the integral is 0. $\Rightarrow I = \mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^4} = 0$

Exercise 4.3

$$A_0(m^2) = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^2 - m^2}$$

$$B_0(m_1^2, m_2^2, p^2) = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 - m_1^2)((k-p)^2 - m_2^2)}$$

$$C_0(m_1^2, m_2^2, m_3^2, p_1^2, p_2^2) = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)}$$

Reduce

$$C^\mu = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{k^\mu}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)}$$

C^μ only has one Lorentz index and it's a function of p_1 and p_2

$\Rightarrow$ It's Lorentz structure must be of following form:

$$C^\mu = p_1^\mu C_1 + p_2^\mu C_2$$

Let's now use projection to figure out C_1 & C_2

$$\begin{cases} p_{1\mu} C^\mu = p_{1\mu} p_1^\mu C_1 + p_{1\mu} p_2^\mu C_2 = p_1^2 C_1 + (p_1 \cdot p_2) C_2 \end{cases} \quad (1)$$

$$\begin{cases} p_{2\mu} C^\mu = p_{2\mu} p_1^\mu C_1 + p_{2\mu} p_2^\mu C_2 = (p_1 \cdot p_2) C_1 + p_2^2 C_2 \end{cases} \quad (2)$$

Let's investigate eq. (1):

$$p_{1\mu} C^\mu = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{p_1 \cdot k}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)} = p_1^2 C_1 + (p_1 \cdot p_2) C_2$$

Let's now try to get our nominator into form independent of k via algebra

$$(k-p_1)^2 - m_2^2 = (k^2 - m_1^2) + m_1^2 - 2k \cdot p_1 + p_1^2 - m_2^2$$

$$2k \cdot p_1 = (k^2 - m_1^2) - ((k-p_1)^2 - m_2^2) + p_1^2 + m_1^2 - m_2^2$$

$$\Rightarrow p_{1\mu} C^\mu = -\frac{i}{2} \mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \left[\frac{1}{((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)} - \frac{1}{(k^2 - m_1^2)((k-p_1-p_2)^2 - m_3^2)} + \frac{p_1^2 + m_1^2 - m_2^2}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)} \right]$$

We can recognise the scalar integrals B_0 and C_0

$$p_{1\mu} C^\mu = \frac{1}{2} B_0(m_2^2, m_3^2, p_2^2) - \frac{1}{2} B_0(m_1^2, m_3^2, (p_1+p_2)^2) + \frac{p_1^2 + m_1^2 - m_2^2}{2} C_0(m_1^2, m_2^2, m_3^2, p_1^2, p_2^2)$$

Note that in the first B_0 we have only p_2^2 and this is fine since we can do change of variables: $k \rightarrow k-p_1 = \ell \Rightarrow \int d^D k \frac{1}{((k-p_1)^2 - m^2)} \rightarrow \int d^D \ell \frac{1}{(\ell^2 - m^2)}$ since $dk = d\ell$

Now let's investigate $P_{2\mu} C^\mu$

$$P_{2\mu} C^\mu = -i\mu^\varepsilon \int \frac{d^D k}{(2\pi)^D} \frac{P_2 \cdot k}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)}$$

Algebra: $(k-p_1-p_2)^2 = (k-p_1)^2 - m_2^2 + m_2^2 - 2P_2 \cdot (k-p_1) + P_2^2$

$$(k-p_1-p_2)^2 = [(k-p_1)^2 - m_2^2] - 2P_2 \cdot k - 2P_1 \cdot P_2 + P_2^2 + m_2^2$$

$$\Rightarrow 2P_2 \cdot k = [(k-p_1)^2 - m_2^2] - [(k-p_1-p_2)^2 - m_3^2] - 2P_1 \cdot P_2 + P_2^2 + m_2^2 - m_3^2$$

$$\Rightarrow P_{2\mu} C^\mu = -\frac{i}{2}\mu^\varepsilon \int \frac{d^D k}{(2\pi)^D} \left[\frac{1}{(k^2 - m_1^2)((k-p_1-p_2)^2 - m_3^2)} - \frac{1}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)} + \frac{-2P_1 \cdot P_2 + P_2^2 + m_2^2 - m_3^2}{(k^2 - m_1^2)((k-p_1)^2 - m_2^2)((k-p_1-p_2)^2 - m_3^2)} \right]$$

We recognise B_0 and C_0 Integrals

$$\Rightarrow P_{2\mu} C^\mu = \frac{1}{2} B_0(m_1^2, m_3^2, (P_1+P_2)^2) - \frac{1}{2} B_0(m_1^2, m_2^2, P_1^2) + \frac{P_1^2 - 2P_1 \cdot P_2 + m_2^2 - m_3^2}{2} C_0(m_1^2, m_2^2, m_3^2, P_1^2, P_2^2)$$

Now we go back to the equation pair (1) & (2) to get C_1 and C_2

$$\begin{cases} P_{1\mu} C^\mu = P_1^2 C_1 + (P_1 \cdot P_2) C_2 = \frac{1}{2} B_0(m_2^2, m_3^2, P_2^2) - \frac{1}{2} B_0(m_1^2, m_3^2, (P_1+P_2)^2) + \frac{P_1^2 + m_1^2 - m_2^2}{2} C_0(m_1^2, m_2^2, m_3^2, P_1^2, P_2^2) \\ P_{2\mu} C^\mu = (P_1 \cdot P_2) C_1 + P_2^2 C_2 = \frac{1}{2} B_0(m_1^2, m_3^2, (P_1+P_2)^2) - \frac{1}{2} B_0(m_1^2, m_2^2, P_1^2) + \frac{P_1^2 - 2P_1 \cdot P_2 - m_2^2 - m_3^2}{2} C_0(m_1^2, m_2^2, m_3^2, P_1^2, P_2^2) \end{cases}$$

$$\Rightarrow P_1^2 C_1 = P_{1\mu} C^\mu - (P_1 \cdot P_2) C_2 \quad \& \quad C_2 = \frac{1}{P_2^2} [P_{2\mu} C^\mu - (P_1 \cdot P_2) C_1]$$

Let's write these as just C_0 since we encounter it with same arguments

$$\Rightarrow P_1^2 C_1 = P_{1\mu} C^\mu - \frac{P_1 \cdot P_2}{P_2^2} [P_{2\mu} C^\mu - (P_1 \cdot P_2) C_1] = P_{1\mu} C^\mu - \frac{P_1 \cdot P_2}{P_2^2} P_{2\mu} C^\mu + \frac{(P_1 \cdot P_2)^2}{P_2^2} C_1$$

$$C_1 \left[P_1^2 + \frac{(P_1 \cdot P_2)^2}{P_2^2} \right] = P_{1\mu} C^\mu - \frac{P_1 \cdot P_2}{P_2^2} P_{2\mu} C^\mu$$

$$\Rightarrow C_1 = \frac{P_2^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} \left[\frac{1}{2} B_0(m_2^2, m_3^2, P_2^2) - \frac{1}{2} B_0(m_1^2, m_3^2, (P_1+P_2)^2) + \frac{P_1^2 + m_1^2 - m_2^2}{2} C_0 - \frac{P_1 \cdot P_2}{P_2^2} \left(\frac{1}{2} B_0(m_1^2, m_3^2, (P_1+P_2)^2) - \frac{1}{2} B_0(m_1^2, m_2^2, P_1^2) + \frac{P_1^2 - 2P_1 \cdot P_2 - m_2^2 - m_3^2}{2} C_0 \right) \right]$$

$$\Rightarrow C_1 = \frac{P_2^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} \cdot \frac{1}{2} \left(B_0(m_1^2, m_3^2, (P_1+P_2)^2) \left(-1 - \frac{P_1 \cdot P_2}{P_2^2} \right) + C_0 \left[P_1^2 + m_1^2 - m_2^2 - \frac{P_1 \cdot P_2}{P_2^2} (P_1^2 - 2P_1 \cdot P_2 - m_2^2 - m_3^2) \right] + B_0(m_2^2, m_3^2, P_2^2) + \frac{P_1 \cdot P_2}{P_2^2} B_0(m_1^2, m_2^2, P_1^2) \right)$$

$$\Rightarrow \underline{C_1} = -\frac{1}{2} \frac{P_2^2 + P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, (P_1 + P_2)^2) + \frac{1}{2} \frac{P_1^2 P_2^2 + m_1^2 P_2^2 - m_3^2 P_2^2 - P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} \cdot (P_1^2 - 2(P_1 \cdot P_2) - m_2^2 - m_3^2) \cdot C_0$$

$$+ \frac{1}{2} \frac{P_2^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, P_2^2) + \frac{1}{2} \frac{P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_2^2, P_1^2)$$

Now let's calculate $C_2 = \frac{1}{P_2^2} [P_{1\mu} C_1 - (P_1 \cdot P_2) C_1]$

$$\Rightarrow C_2 = \frac{1}{2P_2^2} \left[B_0(m_1^2, m_3^2, (P_1 + P_2)^2) - B_0(m_1^2, m_2^2, P_1^2) + (P_1^2 - 2(P_1 \cdot P_2) - m_2^2 - m_3^2) C_0 \right.$$

$$- (P_1 \cdot P_2) \left[\frac{P_2^2 + (P_1 \cdot P_2)}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, (P_1 + P_2)^2) + \frac{P_1^2 P_2^2 + m_1^2 P_2^2 - m_2^2 P_2^2 - P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} (P_1^2 - 2(P_1 \cdot P_2) - m_2^2 - m_3^2) \cdot C_0 \right.$$

$$\left. + \frac{P_2^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, P_2^2) + \frac{P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_2^2, P_1^2) \right]$$

$$C_2 = \frac{1}{2P_2^2} \left[B_0(m_1^2, m_3^2, (P_1 + P_2)^2) \left[1 - \frac{P_2^2 (P_1 \cdot P_2) + (P_1 \cdot P_2)^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} \right] + B_0(m_1^2, m_2^2, P_1^2) \left[-1 - \frac{(P_1 \cdot P_2)^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} \right] \right.$$

$$+ C_0 \cdot \left[P_1^2 - 2(P_1 \cdot P_2) - m_2^2 - m_3^2 - (P_1 \cdot P_2) \frac{P_1^2 P_2^2 + m_1^2 P_2^2 - m_2^2 P_2^2 - P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} (P_1^2 - 2(P_1 \cdot P_2) - m_2^2 - m_3^2) \right.$$

$$\left. - \frac{P_2^2 (P_1 \cdot P_2)}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, P_2^2) \right]$$

$$\underline{C_2} = \frac{1}{2} \frac{P_1^2 - P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, (P_1 + P_2)^2) + \frac{1}{2P_2^2} \frac{P_1^2 P_2^2 + 2(P_1 \cdot P_2)^2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_2^2, P_1^2)$$

$$- \frac{1}{2} \frac{P_1 \cdot P_2}{P_1^2 P_2^2 + (P_1 \cdot P_2)^2} B_0(m_1^2, m_3^2, P_2^2)$$

$$+ \frac{1}{2} \frac{(P_1^2 - 2P_1 \cdot P_2 - m_2^2 - m_3^2)(P_1^2 P_2^2 + 2(P_1 \cdot P_2)^2 + P_2^2 (P_1 \cdot P_2) [m_2^2 - m_1^2 - P_1^2])}{P_2^2 (P_1^2 P_2^2 + (P_1 \cdot P_2)^2)} C_0(m_1^2, m_2^2, m_3^2, P_1^2, P_2^2)$$

With the results we got for C_1 and C_2 we have reduced the integral to scalar integrals with

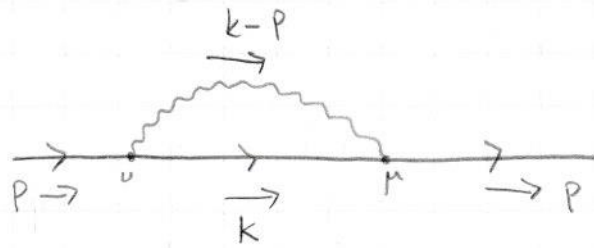
$$C^\mu = P_1^\mu C_1 + P_2^\mu C_2$$

with full expression for C_1 and C_2 above.

C_1 and C_2 are functions of B_0 and C_0 which are scalar integrals so the reduction is truly into a scalar integral.

Exercise 4.4

Electron self energy :



The diagram gives the following integral in dimensional regularization :

$$I = \mu^\epsilon \int \frac{d^D k}{(2\pi)^D} (-ie\gamma^\nu) \frac{i(k+m)}{(k^2-m^2+i\epsilon)} (-ie\gamma^\mu) \frac{-i g_{\mu\nu}}{(k-p)^2 + i\epsilon}$$

Let's suppress the $i\epsilon$ -prescription and contract with $g_{\mu\nu}$ and using $\gamma^\nu \gamma_\nu = D$

$$I = -\mu^\epsilon e^2 \int \frac{d^D k}{(2\pi)^D} \frac{\gamma^\nu (k+m) \gamma_\nu}{(k^2-m^2)((k-p)^2)} = -e^2 \mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \left[\frac{\gamma^\nu k \gamma_\nu}{(k^2-m^2)(k-p)^2} + \frac{D m}{(k^2-m^2)((k-p)^2 - 0^2)} \right]$$

$\uparrow$
Zero

We can recognise the $B_0(m^2, 0^2, p^2) = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2-m^2)((k-p)^2 - 0^2)}$

$$\Rightarrow I = -e^2 \mu^\epsilon \left[\int \frac{d^D k}{(2\pi)^D} \frac{\gamma^\nu k \gamma_\nu}{(k^2-m^2)(k-p)^2} \right] - ie^2 B_0(m^2, 0, p^2)$$

$= \mathcal{L}$

Let's now calculate $\mathcal{L} = \int \frac{d^D k}{(2\pi)^D} \frac{\gamma^\nu k \gamma_\nu}{(k^2-m^2)(k-p)^2}$ || Peskin (A.55) $\gamma^\nu \gamma^\rho \gamma_\nu = -(D-2)\gamma^\rho$ from

$$\Rightarrow \mathcal{L} = \int \frac{d^D k}{(2\pi)^D} \frac{-(D-2) k}{(k^2-m^2)(k-p)^2} = -(D-2)\gamma^\mu \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu}{(k^2-m^2)(k-p)^2}$$

Let's now use the P-V reduction integral (Minkowski space definition) :

$$B_\mu(m_1^2, m_2^2, p^2) = -i\mu^\epsilon \int \frac{d^D k}{(2\pi)^D} \frac{k_\mu}{(k^2-m_1^2)((k-p)^2 - m_2^2)}$$

$$\Rightarrow I = +ie^2 (D-2)\gamma^\mu B_\mu(m^2, 0, p^2) - ie^2 B_0(m^2, 0, p^2)$$

And now we use $B_\mu(m_1^2, m_2^2, p^2) = P_\mu B_1 = P_\mu \frac{1}{2p^2} (A_0(m_1^2) - A_0(m_2^2) - (p^2 + m_1^2 - m_2^2) B_0(m_1^2, m_2^2, p^2))$

$$\Rightarrow I = ie^2 (D-2) \cancel{P} \frac{1}{2p^2} [A_0(m^2) - A_0(0) - (p^2 + m^2) B_0(m^2, 0, p^2)] - ie^2 B_0(m^2, 0, p^2)$$

$\stackrel{\sim}{=} 0$, Since no external scale in integral (according to Exercise 4.2)

$$\Rightarrow I = \frac{ie^2 (D-2)}{2 p^2} \cancel{P} A_0(m^2) - ie^2 \left[\frac{(D-2) \cancel{P}}{2 p^2} (p^2 + m^2) + 1 \right] B_0(m^2, 0, p^2)$$

This is the expression for electron self-energy